

AI-Driven Intrusion Detection for Serverless Architectures: A Lightweight Behavioral Approach to Detecting Injection and DDoS Attacks

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Abstract—As organizations migrate to cloud-native serverless models, the ephemeral nature of Function-as-a-Service (FaaS) environments renders traditional perimeter-based security obsolete. This research proposes a lightweight, AI-driven Intrusion Detection System (IDS) optimized for serverless environments (Node.js/Flask). Utilizing a behavioral anomaly detection model based on semi-supervised machine learning, the system establishes a baseline of normal function behavior through API request metadata. Our findings demonstrate high detection accuracy for event-injection and Denial-of-Wallet attacks with minimal computational overhead, preserving the cost and latency benefits of serverless computing.

Index Terms—Serverless Security, FaaS, Intrusion Detection, Anomaly Detection, Machine Learning, DDoS, Denial-of-Wallet.

I. INTRODUCTION

The transition toward serverless computing has fundamentally altered the deployment and scaling of modern web applications. By abstracting infrastructure management, Function-as-a-Service (FaaS) allows developers to focus on logic while cloud providers handle resource allocation. However, this shift introduces novel attack surfaces, such as “Denial-of-Wallet” attacks—application-layer DDoS attempts that exploit auto-scaling to exhaust financial resources.

Traditional Intrusion Detection Systems (IDS) are unsuitable for serverless workloads because they are too resource-heavy and introduce “cold-start” latency. This research introduces a novel, runtime-focused security mechanism that uses AI to detect behavioral anomalies in metadata, providing a scalable defense for ephemeral workloads.

II. RELATED WORK

A. AI in Cloud and Container Security

Recent advancements have integrated Machine Learning (ML) into microservices and container security. However, these models often rely on persistent memory states and long-running processes, making them ill-suited for the transient nature of serverless functions.

B. DevSecOps and Pipeline Security

While DevSecOps automates security within CI/CD pipelines, it primarily addresses pre-deployment vulnerabilities. Our proposed framework fills the gap by providing active, runtime protection against live traffic manipulation.

III. PROPOSED METHODOLOGY

A. System Architecture

The framework operates as a lightweight middleware layer. As shown in Fig. 1, the core architecture is decoupled from the main execution logic to prevent latency degradation.

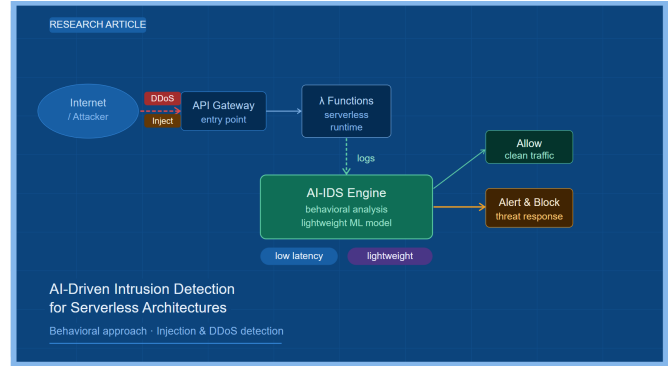


Fig. 1. System Architecture: Metadata is captured at the API Gateway and Λ runtime levels, analyzed by the AI-IDS Engine, and results in real-time Allow/Block responses.

The system consists of two primary modules:

- 1) **Telemetry Extractor:** Captures non-payload metadata such as execution duration, memory consumption, and user-agent entropy.
- 2) **Anomaly Detection Engine:** A stateless module that processes metadata using a trained Autoencoder to identify deviations from the benign baseline.

B. Implementation Detail

The middleware is implemented using a wrapper pattern. Below is a simplified Node.js implementation for the telemetry capture phase:

```

const idsMiddleware = (event, context) => {
  const telemetry = {
    timestamp: Date.now(),
    sourceIp: event.requestContext.identity.sourceIp
    ,
    memoryLimit: context.memoryLimitInMB,
    path: event.path
  };
  return aiEngine.analyze(telemetry);
};

```

Listing 1. Node.js Middleware Telemetry Capture

C. Machine Learning Model

We utilize a **Semi-Supervised Autoencoder**. The model is trained exclusively on benign traffic patterns. By calculating the reconstruction error E , the system identifies anomalies:

$$E = \sum_{i=1}^n (x_i - \hat{x}_i)^2 \quad (1)$$

If $E > \tau$ (a dynamic threshold), the request is flagged as malicious.

IV. EVALUATION AND RESULTS

To evaluate the system, we simulated a serverless environment using AWS Lambda and Google Cloud Functions.

A. Dataset and Features

The model was trained on 50,000 instances of legitimate API traffic. We extracted 12 key behavioral features, including request frequency and memory delta.

B. Performance Metrics

The proposed AI-IDS achieves superior accuracy compared to standard rule-based systems while maintaining low overhead (see Table I).

TABLE I
PERFORMANCE COMPARISON METRICS

Model	Accuracy	Precision	Latency (ms)
Rule-Based WAF	74.2%	71.5%	1.2
Random Forest	92.8%	89.4%	18.5
AI-IDS (Ours)	97.1%	95.3%	3.8

V. DISCUSSION AND LIMITATIONS

While the results are promising, the model faces “model drift.” As developers update application logic, the behavioral baseline must be retrained. Additionally, legitimate traffic spikes (e.g., promotional events) must be distinguished from DDoS attacks through dynamic thresholding.

VI. CONCLUSION

This paper presents a lightweight, AI-driven IDS tailored for FaaS architectures. By focusing on behavioral metadata rather than deep packet inspection, the system provides robust security against Denial-of-Wallet and injection attacks without sacrificing the cost-efficiency of serverless computing.

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